

PYLON low voltage Protocol RS485

Version 3.5

Date	Version	Chapter	Note
2008/11/20	V2.2		初始版本
2008/12/23	V2.3		1. 获取模块数量，相应的数据位置更改 2. 获取系统参数内数据，增加单体电池欠压门限和总电压欠压门限 3. 0x42, 0x44 命令中增加对 Command 不为 0xFF 的定义 4. 警告信息内的 State2 数据增加：使用 Pack 供电指示
2009/03/26	V2.4		1. *命令格式内 ADR 都是指主机地址 2. 增加对 Buzzer 功能开关的命令—3.8 3. Buzzer 功能的指示—State3 的 bit0
2009/12/07	V2.4		1. 在告警数据中增加两个字节 State4 和 State5，用来表示单个电池故障 2. 告警数据中字节 State3 的 bit6（有效放电电流指示）作为交流停电告警来解释
2016/06/20	V2.5		1. 修改“4 说明”中关于电流单位的描述，统一修改为实际值 = 传递值 * 100 2. 增加“5 例程”实例解析 3. 增加通讯接口和传输速率的描述 4. 删除不支持的命令
2016/08/19	V2.6		1. 增加命令“获取充放电管理信息”和“获取序列号信息”

			Add command: get information of charge and discharge information; get SN number
2016/10/13	V2.7		1. 增加命令“设置充放电管理信息” Add command: set charge and discharge parameter
2016/12/15	V2.8		1. 增加命令“关机” Add command: turn off
2017/01/17	V2.9		1. 增加命令“获取软件版本” Add command: get firmware version
2017/11/22	V2.9		优化翻译
2018/03/08	V3.0	2.5.3 3.5-status1-bit1 3.3; 3.4; 3.5;	1. 增补多组并联模式，扩展地址数量 Add multi group mode 2. 更正：单芯低压改为欠压 Correct cell low voltage to cell undervoltage 3. 删除章节 4，将内容移至每个模拟量的 note 里，方便解析阅读 Delete old chapter 4, move all note info to each analog data.
2018/04/08	V3.1	3.6	1. 增加强充请求标志 2，增加满充请求 Add charge immediately 2, add full charge request
2018/06/04	V3.2	3.3 4 3.6 3.5	1. Add item to send capacity of battery bigger than 65Ah. 2. Modified examples 3. Add info to help reading.
2018/08/21	V3.3	3.6	1. 增补 bit 含义说明 Add instruction of bit 6~ bit 7
2018/09/27	V3.4	3.2 3.10	1. 增补阅读说明 Add instruction for reading 2. 修正描述错误 Modified wrong description
2019/12/23	V3.5		Three part: standard RS485 protocol, system info by RS485, upgrade function by RS485. 分为三部分。标准信息，系统信息，升级功能。 Note: for UP2500 series, the 51H, 92H, 42H is special designed. The firmware, module, and protocol lookup table. Please ask us for detailed info. 协议、产品、软件的对应关系，详询技术支持人员。 Modified wrong description in 2.3.3

1 目录

1. 通讯协议说明/PROTOCOL INFO	4
2. 通信协议/PROTOCOL	4
2.1 基本格式/Basic format.....	4
2.1.1 帧基本格式/Basic format of frame	4
2.1.2 帧说明/Introduction of frame	5
2.2 数据格式/Data format	6
2.2.1 Basic data format	6
2.2.2 LENGTH data format	6
2.2.3 CHKSUM data format	7
2.2.4 DATA INFO data format	7
2.2.5 DATA TIME and COMMAND TIME format	7
2.3 Module introduction	8
2.4 编码表/Encoding table	8
2.4.1 CID1.....	8
2.4.2 CID2.....	8
2.4.3 ADR settings / 地址设置.....	10
3. 逐台查询命令/COMMUNICATION COMMAND FOR SINGLE GROUP	11
3.1 获取协议版本号/get protocol version	11
3.2 获取厂商信息/get manufactory info	11
3.3 获取模拟量量化数据, 定点型/get analog value, fixed point.....	12
3.4 获取系统参数, 定点数/get system parameter, fixed point	14
3.5 获取告警信息/get alarm info.....	15
3.6 获取电池充放电管理信息/get charge, discharge management info.....	18
3.7 获取设备序列号/get module SN number.....	19
3.8 设定充放电管理信息/setting charge/discharge management info	20
3.9 关机/turn off module	21
3.10 获取软件版本/get software version	21
4. 系统查询命令/COMMUNICATION COMMAND FOR SYSTEM INFO	23
4.1 系统基本信息/Basic system information.....	23
4.2 系统模拟量/Analog data of system.....	23
4.3 系统告警保护/alarm & protection info of system	27
4.4 系统交互信息 / system recommend info	28
5. 升级功能 / UPGRADE FUNCTION.....	29

1. 通讯协议说明/Protocol info

当电池与上位机通信时，默认上位机为主机，第一台电池是从机，地址/ADR 从 2 开始
When battery communicate with inverter, the first Address of battery starts from 2

拨码开关的设置，详见产品规格书。
Setting of dip switch, please refer to specification of product.

当单组模式时。主机拨码开关默认 X000。
When in single group mode, master battery dip address usually is X000.

当多组级联模式时。主机地址拨码开关需要正确配置。XXXX
When in multi-group mode, master battery shall have right dip switch address.

模块/电池模块/Module/Battery = battery module with BMS
电芯/Cell = 3.2V cell

2. 通信协议/Protocol

Transmission rate: RS485: 115.2kb/s (recommend), 9.6kb/s
通信传输格式为：起始位 1 位，数据位 8 位，停止位 1 位，无校验。
Format: Start bit 1 bit, Data bit 8 bit, Stop bit 1 bit, Without parity

2.1 基本格式/Basic format

2.1.1 帧基本格式/Basic format of frame

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
Format	SOI	VER	ADR	CID1	CID2	LENGTH	INFO	CHKSUM	EOI

2.1.2 帧说明/Introduction of frame

No	Mark	Meaning	
1	SOI	起始位标志/Start bit mark	7EH
2	VER	协议版本号/Version of protocol	Fixed value 32 30
3	ADR	地址 (0、255 保留) /Address	Single pile: start from 2
4	CID1	控制标识码/Control identify code	46H
5	CID2	命令信息：控制标识码 Command information: 应答信息：返回码 Response information: return code	
6	LENGTH	INFO 字节长度，包括 LENID 和 LCHKSUM INFO length, including LENID and LCHKSUM	
7	INFO	命令信息：控制数据信息 Command information: command INFO 应答信息：应答数据信息 Response information: data INFO Flag, data INFO	
8	CHKSUM	校验码/CHECKSUM	
9	EOI	结束码/End code	0DH

■ Command INFO

Command group	1 byte	同一类型设备的不同组号 Group number of same type of device
Command type	1 byte	不同的遥控命令；历史数据传输中的不同控制命令 Different remote control command or different control command in history data transmission
Command id	1 byte	同一类型设备相同组内的不同监控点 Different monitoring point of same type device group
Command time	7 bytes	时间字段 Time field, see table data time format

■ Data INFO

DATAI	定点数应答信息（本协议不采用定点数） Fixed point number response information (not included in this protocol)
DATAF	浮点数应答信息 Floating point number response information
DATA FLAG	数据标识信息 Data flag information
RUN STATE	设备运行状态 Status of battery
WARN STATE	设备告警状态 Alarm information

■ Data INFO flag format

	Bit 7	Bit 6	Bit 5	Bit 4		Bit 3	Bit 2	Bit 1	Bit 0	
Value	0	0	0	0	1	0	0	0	0	1
statement				无未读取的 开关量变化 No unread switching value change	有未读取的 开关量变化 Exist unread switching value change				无未读取的 告警量变化 No unread alarm value change	有未读取的 告警量变化 Exist unread alarm value change

2.2 数据格式/Data format

2.2.1 Basic data format

除 SOI 和 EOI 是以 16 进制解释 16 进制传输外，其余各项都是以 16 进制解释，以 16 进制 – ASCII 码方式传输，每个字节用两个 ASCII 码表示，如 4BH，传输时传送 34H（‘4’的 ASCII 码）和 42H（‘B’的 ASCII 码）两个字节。

SOI and EOI are explained and transferred in HEX. Other items are explained in HEX, transferred in HEX-ASCII, each byte contains 2 ASCII.

E.g. 4BH, transfer in 2 byte, 34H (“4” in ASCII), and 42H (“B” in ASCII).

2.2.2 LENGTH data format

高字节 HIGH								低字节 LOW							
校验码 CLHKSUM				LENID											
D15	D14	D13	D12	D11	D10	D9	D8	D7	D6	D5	D4	D3	D2	D1	D0

LENID 表示 INFO 项的 ASCII 码字节数，当 LENID = 0 时，INFO 为空，即无该项。

LENID means the number of byte of ASCII in INFO, when LENID = 0, means INFO is empty.

由于 LENID 只有 12Bit，所以，要求数据包最大不能超过 4095 个字节。

LENID has 12 bits, data package should smaller than 4095 bytes.

LENGTH 传输中先传高字节，再传低字节，分四个 ASCII 码传送。

While transmission, HIGH byte first, then LOW byte and divided into 4 ASCII to transmit.

校验码 LCHKSUM 的计算：D11D10D9D8+D7D6D5D4+D3D2D1D0，求和后模 16 余数 取反加 1

To calculate LCHKSUM: D11D10D9D8+D7D6D5D4+D3D2D1D0, add the sum, modulus 16 take remainder, then do a bitwise invert and then plus 1.

e.g.:

INFO 中 ASCII 码字节数为 18，即 LENID = 00000010010B。

In INFO the number of ASCII is 18, then LENID = 000000010010B
D11D10D9D8+D7D6D5D4+D3D2D1D0 = 0000B + 0001B + 0010B = 0011B,

模 16 余数为/ modulus 16 the remainder = 0011B,
取反加 1 为/do a bitwise invert and plus 1 = 1101B,

LCHKSUM = 1101B。

LENGTH = 1101000000010010B, trans: D012

2.2.3 CHKSUM data format

CHKSUM 的计算是除 SOI、EOI 和 CHKSUM 外，其他字符按 ASCII 码值累加求和，所得结果模 65536 余数取反加 1。

Except for SOI, EOI and CHKSUM, add sum number of other characters in ASCII, the result modulus 65535 take remainder, then do a bitwise invert and then plus 1.

E.g.:

收到或发送的字符序列是：“~1203400456ABCEFEFC71\R” (“~”为 SOI, “CR”为 EOI),

If we have a character: “~1203400456ABCEFEFC71\R ” (“~” is SOI, “CR” is EOI)

则最后 5 个字符“FC71\R”中的 FC71 是 CHKSUM,

The last 5 character “FC71\R”, the FC71 is the CHKSUM

Calculate:

‘1’ + ‘2’ + ‘0’ + ... + ‘F’ + ‘E’ = 31H + 32H + 30H + ... + 46H + 45H = 038FH

038FH 模 65535 余码是 038FH, 038FH 取反加 1 就是 FC71H。

038FH modulus 65535 remainder = 038FH, do a bitwise invert and plus 1 = FC71H.

2.2.4 DATA INFO data format

模拟量数据的传送采用定点数和浮点数两种形式，可任选一种。

Analog quantity is transmitted in form of fixed-point or floating-point.

Fixed-point (integer, 2 bytes),

本协议采用定点数/this protocol uses fixed-point

有符号整数/ signed integer: -32768 ~ +32767

无符号整数/ unsigned integer: 0 ~ +65535

2.2.5 DATA TIME and COMMAND TIME format

Year	1-9999	Integer	2 bytes, HEX
Month	1-12	Char	1 byte, HEX
Day	1-31	Char	1 byte, HEX
Hour	0-23	Char	1 byte, HEX

Minute	0-59	Char	1 byte, HEX
Second	0-59	Char	1 byte, HEX
Note	Year is transferred as integer, actual value = transfer value		

2.3 Module introduction

RS485 总线形式通信，默认速率 115200 BPS

Use RS485 bus with default communication rate 115200 BPS.

2.4 编码表/Encoding table

2.4.1 CID1

No	Content	CID1	Note
1	电池数据/battery data	46H	

2.4.2 CID2

■ 命令信息/command information

标准内容: 派能 48V, 24V 系列支持逐个查询模式。

standard: point to point communication, supported by 48V and 24V series.

No	Content	CID2	Note
1	获取模拟量数据，定点数 Get analog value, fixed point	42H	电池基本状态 Basic info
2	获取告警 Get alarm info	44H	告警，保护 Alarm, protection
3	获取系统参数，定点数 Get system parameter, fixed point	47H	可以不使用，BMS 保护值 Fixed info. BMS protection value
4	获取通信协议版本 Get protocol version	4FH	可以不使用 Not useful
5	获取厂商信息 Get manufacturer info	51H	型号、品牌，内容固定 本条目下的版本信息可能错误，以 96H 为准 The firmware version in this part is not correct, use 96H instead.
6	获取充放电管理信息 Get charge, discharge management info	92H	运行交互信息 Running info.
7	获取序列号 Get SN number of battery	93H	

8	设置充放电管理信息 Set value of charge, discharge management info	94H	不建议使用 Unrecommended
9	关机 Turnoff	95H	谨慎使用
10	获取软件版本 Get firmware info	96H	固件版本, 内容固定 Firmware version

系统信息/system info: for some products. Ask master battery to get

11	获取电池组系统基本信息 Get system basic information	60H	
12	获取电池组系统运行模拟量数据 Get System analog data	61H	
13	获取电池组系统状态告警量数据 Get system alarm info	62H	
14	获取电池组系统充放电管理交互信息 Get system charge discharge management info	63H	

■ 响应信息/response information

No	Content	CID2	Note
1	Normal	00H	
2	VER error	01H	
3	CHKSUM error	02H	
4	LCHKSUM error	03H	
5	CID2 invalid	04H	
6	Command format error	05H	
7	Invalid data	06H	INFO data invalid
8	ADR error	90H	
9	Communication error	91H	Internal communication error

■ 定点数数据类型/ fixed point type

No	Telemetry content	Data type
1	Cell voltage	Signed integer
2	Temperature	Signed integer
3	Module voltage	Unsigned integer
4	Module current	Signed integer, charge is +
5	System parameter	Signed integer
6	capacity	Unsigned integer

2.4.3 ADR settings / 地址设置

每组最多 N 台级联(参见电池产品说明书), 电池地址为:

Maximum number in one group please refer to product specification, battery address is

(n)	Position		
2	Master battery	C	Slave 10
3	Slave 1	D	Slave 11
4	Slave 2	E	Slave 12
5	Slave 3	F	Slave 13
6	Slave 4	10	Slave 14
7	Slave 5	12	Slave 15
8	Slave 6	13	Slave 16
9	Slave 7	14	Slave 17
A	Slave 8	15	Slave 18
B	Slave 9	16	Slave 19

e.g.: For US2000:

主机的拨码开关第 2-4 位, 用于地址设置规则如下

Dip switch 2-4 of master battery has the function of change the group address.

1 is up, 0 is down

Dip 1	Dip 2	Dip 3	Dip 4	group address (m)
1: RS485 baud rate=9600	0	0	0	0: 单组电池使用时, 请确保主机为 X000, 从机 不做限制/single group only, master battery must follow this setting
0: RS485 baud rate=115200	1	0	0	1: 多组使用时, 第一组需从 X100 开始以确保地 址规则适用/multi-group condition should use this for the first group
Restart to take effect/设 置重启后生 效	0	1	0	2 nd
	1	1	0	3 rd
	0	0	1	4 th
	1	0	1	5 th
	0	1	1	6 th
	1	1	1	7 th

For battery information 获取单台电池的信息的方法

ADR = battery address + group address = 0x0n + 0x10*m

E.g.:

1) Single group slave 4:

n = 5; m = 0

ADR = 0x05 + 0x10*0 = 0x05; INFO of COMMAND = ADR = 0x05

2) multi group, group 3, slave 6;

n = 7; m = 3

ADR = 0x07 + 0x10*3 = 0x37; INFO of COMMAND = ADR = 0x37

3. 逐台查询命令/communication command for single group

3.1 获取协议版本号/get protocol version

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	4FHJ	LENGTH	INFO	CHKSUM	EOI

VER: 任意值/arbitrary value

LENID = 00H

While receiving, battery ignore VER value

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

VER = version

V2.1, VER = 21H

3.2 获取厂商信息/get manufactory info

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	51H	LENGTH	INFO	CHKSUM	EOI

LENID = 0

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

LENID = 40H

■ DATAINFO

No	Content	Data
1	设备名称/battery name	10, integer, ASCII

2	软件版本/software version	2
3	厂商名称/manufacturer name	20, integer, ASCII

3.3 获取模拟量量化数据，定点型/get analog value, fixed point

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	42H	LENGTH	INFO	CHKSUM	EOI

LENID = 02H

INFO is 1 byte = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = INFOFLAG + DATAI

■ DATAI

No	Content	Data
1	Command value	1 byte
2	Data of battery	

➤ Data of battery

No	Content	Data byte	单位及计算说明 Note	精度 Accuracy
1	*电芯节数/number of cell: M	1		
2	Cell 1 voltage	2	V	3
3	Cell 2 voltage	2	V	3
///				
M+1	Cell M voltage	2	V	3
M+2	*温度点数量/number of temperature: N	1		
M+3	1: Temperature of BMS board	2	Kelvin temperature: K Temperature blow 0 the value is negative e.g.: 25.5°C = 25.5*10+2731=2986 -12.4°C = -12.4*10+2731=2607	1
M+4	2: Avg. temperature of cell 1~4	2		
M+5	3: Avg. temperature of cell 5~8	2		
M+6	4: Avg. temperature of cell 9~12	2		
M+7	5: Avg. temperature of cell 13~15/16	2		
///				

M+N+2	Temperature N	2		
M+N+3	Current	2	A Actual value = transmission value * 100 Positive is charge Negative is discharge e.g.: -4000mA = 0xFFD8	3
M+N+4	Module voltage	2	V	3
M+N+5	Remain capacity 1	2	Ah	3
M+N+6	用户自定义个数/ User defined items = 2(battery capacity≤65Ah) User defined items = 4(battery capacity > 65Ah)	1		
M+N+7	Module total capacity 1	2	Ah	3
M+N+8	Cycle number	2		
M+N+9	**Remain capacity 2 (For battery capacity > 65Ah)	3	Ah	3
M+N+10	**Module total capacity 2 (For battery capacity > 65Ah)	3	Ah	3

*For different product, the number might change.

**To be compatible with old version, we add more items, used to show the capacity of battery bigger than 65Ah.

For US2000B/US2000B-Plus, user defined items = 2. And use remain capacity 1 and module total capacity 1.

For US3000 or big capacity (> 65Ah), user defined items = 4, the value: remain capacity 1= FFFF, the module total capacity = FFFF. And please use remain capacity 2, and module total capacity 2.

例子/Example of a 74Ah battery

下发命令/send command:

7E 32 30 30 32 34 36 34 32 45 30 30 32 30 32 46 44 33 33 0D

对应取出数据解析如下表/analysis

7E 32 30 30 32 34 36 30 30 46 30 37 41 31 31 30 32 30 46 30 44 34 35 30 44 34 34 30 44 34 35 30 44 34 34
30 44 34 35 30 44 34 34 30 44 33 45 30 44 34 35 30 44 34 41 30 44 34 41 30 44 34 42 30 44 34 41 30 44 34 41
30 44 34 41 30 44 34 41 30 35 30 42 43 33 30 42 43 33 30 42 43 33 30 42 43 44 30 42 43 44 30 30 30 43 37
32 35 46 46 46 46 30 34 46 46 46 46 30 30 30 32 30 30 43 41 35 38 30 31 32 31 31 30 45 31 41 32 0D

No	content	original data	HEX and decimal	meaning
1	Cell number	30 46	(0FH)15	15 cells
2	Cell1 voltage	30 44 34 35	(0D45H)3397	3397mV
3	Cell2 voltage	30 44 34 34	(0D44H)3396	3396mV

.....		30 44 34 35 30 44 34 34 30 44 34 35 30 44 34 34 30 44 33 45 30 44 34 35 30 44 34 41 30 44 34 41 30 44 34 42 30 44 34 41 30 44 34 41 30 44 34 41		
M+1	CellM voltage	30 44 34 41	(0D4AH)3402	3402mV
M+2	Temperature number N	30 35	(05H)5	5
M+3	温度 1	30 42 43 33	(0BC3H)3011	28 °C
.....		30 42 43 33 30 42 43 33 30 42 43 44		
M+N+2	Temperature N	30 42 43 44	(0BCDH)3021	29 °C
M+N+3	Current	30 30 30 30	(0000H)0	0
M+N+4	Module total voltage	43 37 32 35	(C725H)50981	50981mV
M+N+5	Remain capacity	46 46 46 46	(FFFFH)65535	-
M+N+6	User defined number=4	30 34	(04H)4	4
M+N+7	Module total capacity	46 46 46 46	(FFFFH)65535	-
M+N+8	Cycle number	30 30 30 32	(0002H)2	2
M+N+9	Remain capacity	30 30 43 41 35 38	CA58	51.800Ah
M+N+10	Module total capacity	30 31 32 31 31 30	12110	74.000Ah

3.4 获取系统参数，定点数/get system parameter, fixed point

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	47H	LENGTH	INFO	CHKSUM	EOI

LENID = 0

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = INFOFLAG + DATAI

■ DATAI

No	Content	Data byte	单位及计算 Note	精度 Accuracy
1	单芯高压上限/cell high voltage limit	2	V	3
2	单芯低压/cell low voltage limit	2	V (alarm)	3
3	单芯欠压/cell under voltage limit	2	V (protect)	3
4	充电温度上限/charge high temperature limit	2	K (calculation refer to 3.3)	1
5	充电温度下限/charge low temperature limit	2	K (calculation refer to 3.3)	1
6	充电电流限值/charge current limit	2	A, with symbol, value= transmission value*100	3
7	模块高压上限/module high voltage limit	2	V	3
8	模块低压/module low voltage limit	2	V (alarm)	3
9	模块欠压/module under voltage limit	2	V (protect)	3
10	放电温度上限/discharge high temperature limit	2	K (calculation refer to 3.3)	1
11	放电温度下限/discharge low temperature limit	2	K (calculation refer to 3.3)	1
12	放电电流限值/discharge current limit	2	A, with symbol, value= transmission value*100	3

3.5 获取告警信息/get alarm info

24V 产品的 44H 有变更。

For 24V series, this part has been modified,

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1

format	SOI	VER	ADR	46H	44H	LENGTH	INFO	CHKSUM	EOI
--------	-----	-----	-----	-----	-----	--------	------	--------	-----

LENID = 02H

INFO is 1 byte = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = DATAFLAG + WARNSTATE

■ WARNSTATE

No	Content	Data
1	Command value	1 byte
2	Module alarm info	

➤ Module alarm info

No	Content	Note
1	电芯节数/number of cell: M	1
2	Cell 1 voltage	1
3	Cell 2 voltage	1
///		
M+1	Cell M voltage	1
M+2	*温度点数量/number of temperature: N	1
M+3	BMS Temperature	1
M+4	Cell temperature 1~4	1
M+5	Cell temperature 5~8	1
M+6	Cell temperature 9~12	1
M+7	Cell temperature 13~15/16	1
M+8	MOSFET temperature (for some products)	1
///		
M+N+2	Temperature N	1
M+N+3	Charge current	1
M+N+4	Module voltage	1
M+N+5	Discharge current	1
M+N+6	Status 1	1
M+N+7	Status 2	1
M+N+8	Status 3	1
M+N+9	Status 4	1
M+N+10	Status 5	1

Note. for No. 1 ~ M+N+5

00H: normal

01H: below lower limit (act as protection)

02H: above higher limit (act as protection)

F0H: other error

➤ Status 1

Bit	Content	Note
7	总压欠压/module under voltage: UV	0: normal; 1: trigger
6	充电过温/charge over temperature	0: normal; 1: trigger
5	放电过温/discharge over temperature	0: normal; 1: trigger
4	放电过流/discharge over current: DOC	0: normal; 1: trigger
3		
2	充电过流/charge over current: COC	0: normal; 1: trigger
1	单芯欠压/cell under voltage	0: normal; 1: trigger
0	总压过压/module over voltage: OV	0: normal; 1: trigger

➤ Status 2

Bit	Content	Note
3	使用模块供电/using battery module power	1: using; 0: not
2	Discharge MOSFET	1: on; 0: off
1	Charge MOSFEET	1: on; 0: off
0	Pre MOSFET (for some products)	1: on; 0: off

➤ Status 3

Bit	Content	Note
7	有效充电电流/effective charge current (实际检测到的电流 > 0.1A; current detected by BMS > 0.1A)	1: effective; 0: normal
6	有效放电电流/effective discharge current (实际检测到的电流 < -0.1A; current detected by BMS < -0.1A)	1: effective; 0: normal
5	加热膜启动/heater (reserve, function not suing)	1: on; 0: off
4		
3	充满状态指示/fully charged (SOC=100%)	1: full; 0: normal
2		
1		
0	蜂鸣器功能/buzzer	1: on; 0: off

➤ Status 4

Bit	Content	Note
7	Cell voltage 8	1: error; 0: normal
6	Cell voltage 7	1: error; 0: normal
5	Cell voltage 6	1: error; 0: normal
4	Cell voltage 5	1: error; 0: normal
3	Cell voltage 4	1: error; 0: normal
2	Cell voltage 3	1: error; 0: normal

1	Cell voltage 2	1: error; 0: normal
0	Cell voltage 1	1: error; 0: normal

➤ Status 5

Bit	Content	Note
7	Cell voltage 16	1: error; 0: normal
6	Cell voltage 15	1: error; 0: normal
5	Cell voltage 14	1: error; 0: normal
4	Cell voltage 13	1: error; 0: normal
3	Cell voltage 12	1: error; 0: normal
2	Cell voltage 11	1: error; 0: normal
1	Cell voltage 10	1: error; 0: normal
0	Cell voltage 9	1: error; 0: normal

单芯电压判定为异常。

Error means cell voltage is above threshold value, and supposed to be a failure.

Cell voltage > 4.2V(Battery charge MOS off) or cell voltage < 1V(Battery shut down by itself)

3.6 获取电池充放电管理信息/get charge, discharge management info

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	92H	LENGTH	INFO	CHKSUM	EOI

LENID = 02H

INFO is 1 = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = DATAI

■ DATAI

No	Content	Data
1	Command value	1 byte
2	Charge and discharge management value	

➤ Charge and discharge management value

No	Content	Byte	Note	Accuracy
1	充电电压建议上限/charge voltage limit	2	V	3

2	放电电压建议下限/discharge voltage limit	2	V	3
3	最大充电电流/charge current limit	2	A	1
4	最大放电电流/discharge current limit	2	A	1
5	充放电状态/charge, discharge status	1	-	-

➤ 充放电状态/charge, discharge status

Bit	Content	Note
7	Charge enable	1: normal; 0: request stop charge
6	Discharge enable	1: normal; 0: request stop discharge
5	强充 1, 立即充电/charge immediately	1: request charge; 0: normal
4	强充 2, 立即充电/charge immediately	1: request charge; 0: normal
3	满充请求/full charge request	1: request charge; 0: normal
2		
1		
0		

Bit 5: is SOC 15~19%,

Bit 4: is SOC 9~13%

We suggest inverter to use both charge immediately bit, this two is compatible.

Bit 3:

Reason: when battery is not full charged for long time, the accumulative error of SOC calculation will be too high and may not able to be charged or discharged as expected capacity.

we suggest inverter to charge the battery by grid when this flag is 1.

3.7 获取设备序列号/get module SN number

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	93H	LENGTH	INFO	CHKSUM	EOI

LENID = 02H

INFO is 1 byte = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = DATAI

■ DATAI

No	Content	Data
1	Command value	1 byte
2	Module SN number	16 bytes, integer, ASCII

3.8 设定充放电管理信息/setting charge/discharge management info

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	94H	LENGTH	INFO	CHKSUM	EOI

LENID = 12H

INFO is 9 bytes = command + DataF

Command = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

LENID = 0

➤ DataF

No	Content	Byte	Note	Accuracy
1	充电电压建议上限/charge voltage limit	2		
2	放电电压建议下限/discharge voltage limit	2		
3	最大充电电流/max charge current	2		
4	最大放电电流/min discharge current	2		

NOTE:

需要手动设定充放电管理信息时，按周期发送此命令，电池如果 10 秒未再次收到此命令，电池则按当前条件自动设定充放电管理信息。

未明确 PYLON 关于电流的建议值策略前请勿使用该命令。

Send this command periodically, if the battery doesn't receive this command again in 10 seconds, battery will automatically change setting value using latest command value.

Please contact PYLONTECH first, before doing any change on these settings.

3.9 关机/turn off module

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	95H	LENGTH	INFO	CHKSUM	EOI

LENID = 02H

INFO is 1 byte = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

LENID = 0

3.10 获取软件版本/get software version

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	96H	LENGTH	INFO	CHKSUM	EOI

LENID = 02H

INFO is 1 byte = ADR

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO = DATAI

■ DATAI

No	Content	Data
1	Command value	1 byte
2	Module software version	5 bytes
		2 bytes 3 bytes

		厂商软件版本 Manufacture version	软件主线版本 Main line version
--	--	-------------------------------	-----------------------------

4. 系统查询命令/Communication command for system info

仅用于查询每组电池的主机，获取汇总后的系统信息。

For master battery of each group only. Response the system information.

地址的规则是一致的，当单柜的时候，主机的地址是从 02 开始的。
当多柜的时候，主机的地址是 12，13。。。等等

4.1 系统基本信息/Basic system information

■ command

No	1	2	3	4	5	6	7	8	9
Byte	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	60H	LENGTH	INFO	CHKSUM	EOI

e.g.:

7E 32 30 31 32 34 36 36 30 30 30 30 30 46 44 41 42 0D

■ response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

No	Content	Data
1	主机设备名称/battery name of master module	10, integer, ASCII
2	主机厂商名称/manufacturer name of master module	20, integer, ASCII
3	主机软件版本/software version of master module: AB.CD	2
4	电池的总数量/total number of modules N	1
5	电池 1 的条形码/ SN of master module	16 ASCII
6	电池 2 的条形码/SN of slave 1	16 ASCII
4+N	电池 N 的条形码/SN of slave N-1	16 ASCII

NOTE(2020/02): 本命令/60H 只有 Force L 产品支持，UP2500，US2000，US3000，目前不支持。
Only Force L support this command, UP2500, US2000, US3000 do not support this command.

4.2 系统模拟量/Analog data of system

■ 命令/command

No	1	2	3	4	5	6	7	8	9
----	---	---	---	---	---	---	---	---	---

Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	61H	LENGTH	INFO	CHKSUM	EOI

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

INFO 内容

No	Content	Data byte	单位及计算说明 Note	精度 Accuracy
1	平均电压 / average module voltage	2	V	3
2	总电流 / total current	2	A	2
3	平均 SOC / average SOC	1	%	0
4	平均循环次数 / average cycle number	2	次	0
5	最大循环次数 / max cycle number	2	次	0
6	平均 SOH / average SOH	1	%	0
7	最小 SOH / minimum SOH	1	%	0
8	单芯最高电压 / max cell voltage	2	V	3
9	单芯最高电压的编号最小的模块/ smallest module number of max cell voltage	2	▲: e.g.: 34H: group 3, address 4 (master address = 2) 默认数值: 00/ Default: 00	
10	单芯最低电压 / min cell voltage	2	V	3
11	单芯最低电压的编号最小的模块/ Smallest module number of min cell voltage	2	▲	
12	单芯平均温度 / average cell temperature	2	Kelvin temperature: K Temperature blow 0 the value is negative e.g.: 25.5°C = 25.5*10+2731=2986 -12.4°C = -12.4*10+2731=2607	1
13	单芯最高温度/ max cell temperature	2	Kelvin temperature: K	1
14	单芯最高温度的编号最小的模块 Smallest module number of max cell temperature	2	▲	
15	单芯最低温度 / min cell temperature	2	Kelvin temperature: K	1
16	单芯最低温度的编号最小的模块/ Smallest module number of min cell temperature	2	▲	
17	MOSFET 平均温度 average MOSFET temperature	2	Kelvin temperature: K	1

18	MOSFET 最高温度 Max MOSFET temperature	2	Kelvin temperature: K	1
19	MOSFET 最高温度的编号最小的模块 Smallest module number of max MOSFET temperature	2	▲	
20	MOSFET 最低温度	2	Kelvin temperature: K	1
21	MOSFET 最低温度的编号最小的模块 Smallest module number of min MOSFET temperature	2	▲	
22	BMS 平均温度 / average BMS temperature	2	Kelvin temperature: K	1
23	BMS 最高温度 / max BMS temperature	2	Kelvin temperature: K	1
24	BMS 最高温度的编号最小的模块 Smallest module number of max BMS temperature	2	▲	
25	BMS 最低温度 / min BMS temperature	2	Kelvin temperature: K	1
26	BMS 最低温度的编号最小的模块 Smallest module number of min BMS temperature	2	▲	

NOTE: 若产品不支持某些模拟量的获取，将发送“FF”占位
for any analog information does not support by product, will send FF instead.

e.g.:

1)

7E 32 30 30 32 34 36 36 31 30 30 30 30 46 44 41 42 0D

2) 下发命令/send command:

7E 32 30 31 32 34 36 36 31 30 30 30 30 46 44 41 41 0D

响应数据/receive data:

7E 32 30 31 32 34 36 36 31 38 30 36 32 32 45 35 33 36 31 41 38 36 32 30 39 44 34 30 42 37 34 36 32 36 31 30 44 42 38 30 30 33 34 30 43 42 42 30 30 31 34 30 42 41 41 30 42 42 37 30 30 33 35 30 42 39 44 30 30 31 35 30 42 41 41 30 42 42 38 30 30 33 36 30 42 39 43 30 30 31 36 30 42 41 41 30 42 42 36 30 30 33 37 30 42 39 45 30 30 31 37 45 38 36 32 0D

响应数据含义:

No	Content	Data	
1	电池组系统总平均电压	32 45 35 33	0x2E53, 11.859 V
2	电池组系统总电流	36 31 41 38	0x61A8, 25.00 A
3	电池组系统 SOC	36 32	0x62, 98%
4	平均循环次数	30 39 44 34	0x09D4, 2516
5	最大循环次数	30 42 37 34	0x0B74, 2932
6	平均 SOH	36 32	0x62, 98%
7	最小 SOH	36 31	0x61, 97%
8	单芯最高电压	30 44 42 38	0x0DB8, 3.512 V
9	单芯最高电压所在模块	30 33 30 34	0x0304, group 3+

			address 4
10	单芯最低电压	30 43 42 42	0x0CBB, 3.259 V
11	单芯最低电压所在模块	30 31 30 34	0x0104,
12	单芯平均温度	30 42 41 41	0x0BAA, 25.5 °C
13	单芯最高温度	30 42 42 37	0x0BB7, 26.8 °C
14	单芯最高温度所在模块	30 33 30 35	0x0305,
15	单芯最低温度	30 42 39 44	0x0B9D, 24.2 °C
16	单芯最低温度所在模块	30 31 30 35	0x0105,
17	MOSFET 平均温度	30 42 41 41	0x0BAA, 25.5°C
18	MOSFET 最高温度	30 42 42 38	0x0BB8, 26.9°C
19	MOSFET 最高温度所在模块	30 33 30 36	0x0306,
20	MOSFET 最低温度	30 42 39 43	0x0B9C, 24.1°C
21	MOSFET 最低温度所在模块	30 31 30 36	0x0106,
22	BMS 平均温度	30 42 41 41	0x0BAA, 25.5°C
23	BMS 最高温度	30 42 42 36	0x0BB6, 26.7°C
24	BMS 最高温度所在模块	30 33 30 37	0x0307,
25	BMS 最低温度	30 42 39 45	0x0B9E, 24.3°C
26	BMS 最低温度所在模块	30 31 30 37	0x0107,

4.3 系统告警保护/ alarm & protection info of system

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	62H	LENGTH	INFO	CHKSUM	EOI

e.g.:

7E 32 30 30 32 34 36 36 32 30 30 30 30 46 44 41 41 0D

7E 32 30 31 32 34 36 36 32 30 30 30 30 46 44 41 39 0D

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

No	Content	Byte	Note
1	系统告警状态 1 / alarm 1	1	One or more module has alarm
2	系统告警状态 2 / alarm 2	1	One or more module has alarm
3	系统保护状态 1 / protect 1	1	One or more module under protection
4	系统保护状态 2 / protect 2	1	One or more module under protection

Alarm 1

Bit	Content	Note
7	模块总压高压 / module high voltage	0: normal; 1: trigger
6	模块总压低压 / module low voltage	0: normal; 1: trigger
5	单芯电压高压 / cell high voltage	0: normal; 1: trigger
4	单芯电压低压 / cell low voltage	0: normal; 1: trigger
3	单芯温度高温 / cell high temperature	0: normal; 1: trigger
2	单芯温度低温 / cell low temperature	0: normal; 1: trigger
1	MOSFET 高温 / MOSFET high temperature	0: normal; 1: trigger
0	单芯电压一致性告警 / high voltage difference between cells	0: normal; 1: trigger

Alarm 2

Bit	Content	Note
7	单芯温度一致性告警 / high temperature difference between cells	0: normal; 1: trigger
6	充电过流告警 / charge high current	0: normal; 1: trigger
5	放电过流告警 / discharge high current	0: normal; 1: trigger

4	内部通信错误 / internal communication failure, salve battery offline	0: normal; 1: trigger
3		
2		
1		
0		

Protect 1

Bit	Content	Note
7	模块总压过压 / module over voltage	0: normal; 1: trigger
6	模块总压欠压 / module under voltage	0: normal; 1: trigger
5	单芯电压过压 / cell over voltage	0: normal; 1: trigger
4	单芯电压欠压 / cell under voltage	0: normal; 1: trigger
3	单芯温度过温 / cell over temperature	0: normal; 1: trigger
2	单芯温度欠温 / cell under temperature	0: normal; 1: trigger
1	MOSFET 过温 / MOSFET over temperature	0: normal; 1: trigger
0		

Protect 2

Bit	Content	Note
7		
6	充电过流保护 / charge over current	0: normal; 1: trigger
5	放电过流保护 / discharge over current	0: normal; 1: trigger
4		
3	系统故障保护 / BMS error	0: normal; 1: trigger
2		
1		
0		

4.4 系统交互信息 / system recommend info

■ 命令/command

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	63H	LENGTH	INFO	CHKSUM	EOI

e.g.:

7E 32 30 30 32 34 36 36 33 30 30 30 30 46 44 41 39 0D

7E 32 30 31 32 34 36 36 33 30 30 30 30 46 44 41 38 0D

■ 响应/response

No	1	2	3	4	5	6	7	8	9
Byte number	1	1	1	1	1	2	LENID/2	2	1
format	SOI	VER	ADR	46H	RTN	LENGTH	INFO	CHKSUM	EOI

➤ INFO: Charge and discharge management value

No	Content	Byte	Note	Accuracy
1	充电电压建议上限/charge voltage limit	2	V	3
2	放电电压建议下限/discharge voltage limit	2	V	3
3	最大充电电流/charge current limit	2	A	1
4	最大放电电流/discharge current limit	2	A	1
5	充放电状态/charge, discharge status	1	-	-

➤ 充放电状态/charge, discharge status

Bit	Content	Note
7	Charge enable	1: normal; 0: request stop charge
6	Discharge enable	1: normal; 0: request stop discharge
5	强充, 立即充电/charge immediately	1: request charge; 0: normal
4	满充请求/full charge request	1: request charge; 0: normal
3		
2		
1		
0		

5. 升级功能 / Upgrade function

请洽派能技术支持人员获取升级支持文档。

Please contact us for RS485 upgrade protocol and firmware.